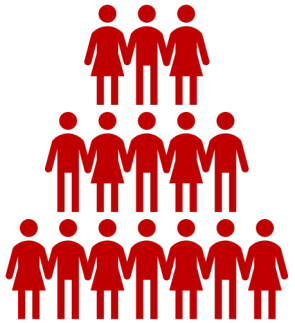
Responsibility Accounting Systems

The key point



A company's owners and top managers must delegate decision-making authority to others except in very SMALL organizations.

When a company's owners (e.g. stockholders) delegate decision-making authority to top managers, they employ corporate governance systems (see lecture #4).



Similarly, when a company's top managers delegate decision-making authority to subordinates, they employ management control systems to direct and control the actions of those subordinates.

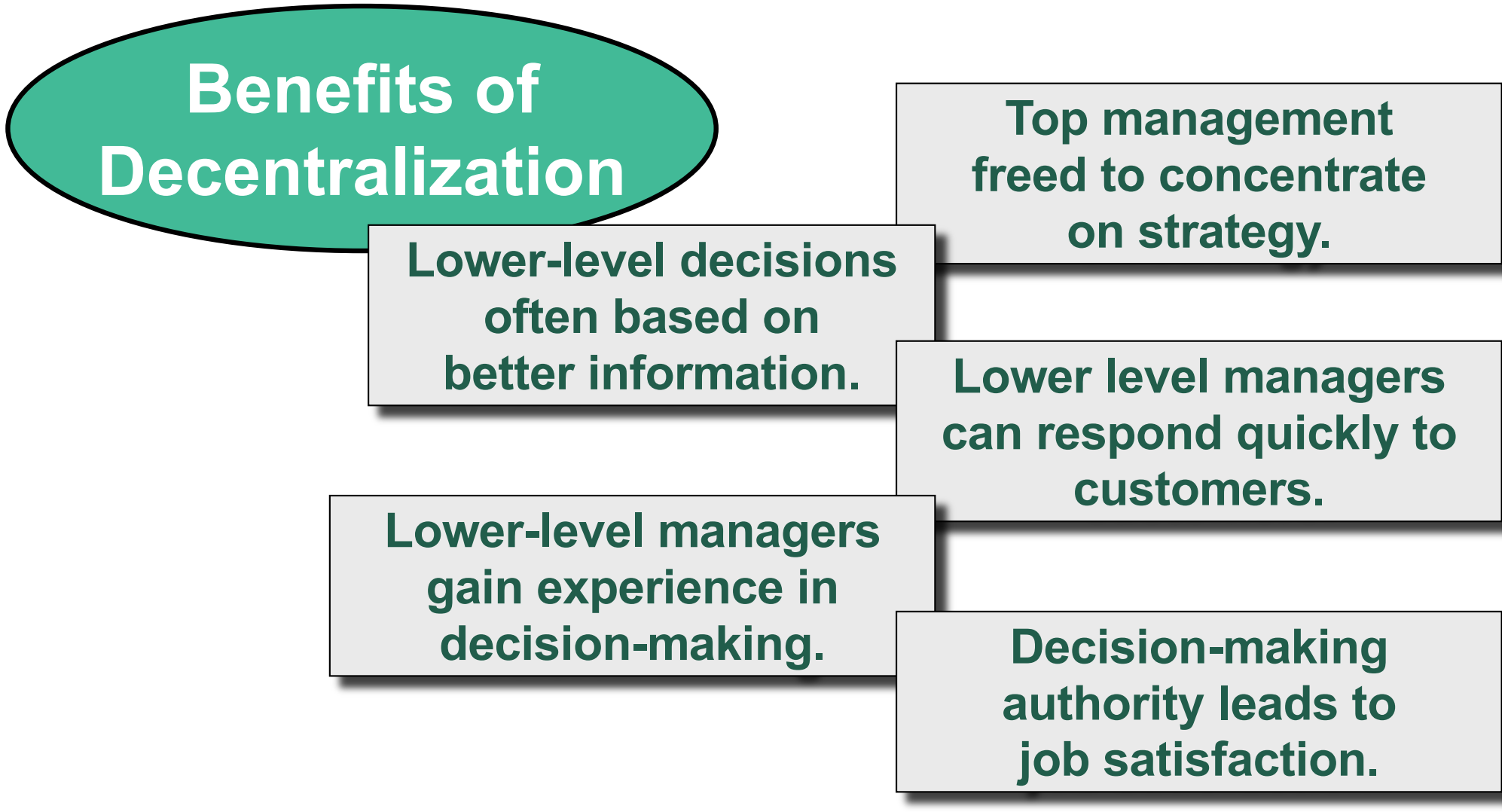
In a DECENTRALIZED ORG, decision-making authoriry is spread throughout the organization rather than being confined to a few top executives.

Decentralization in Organizations: what do we already know?

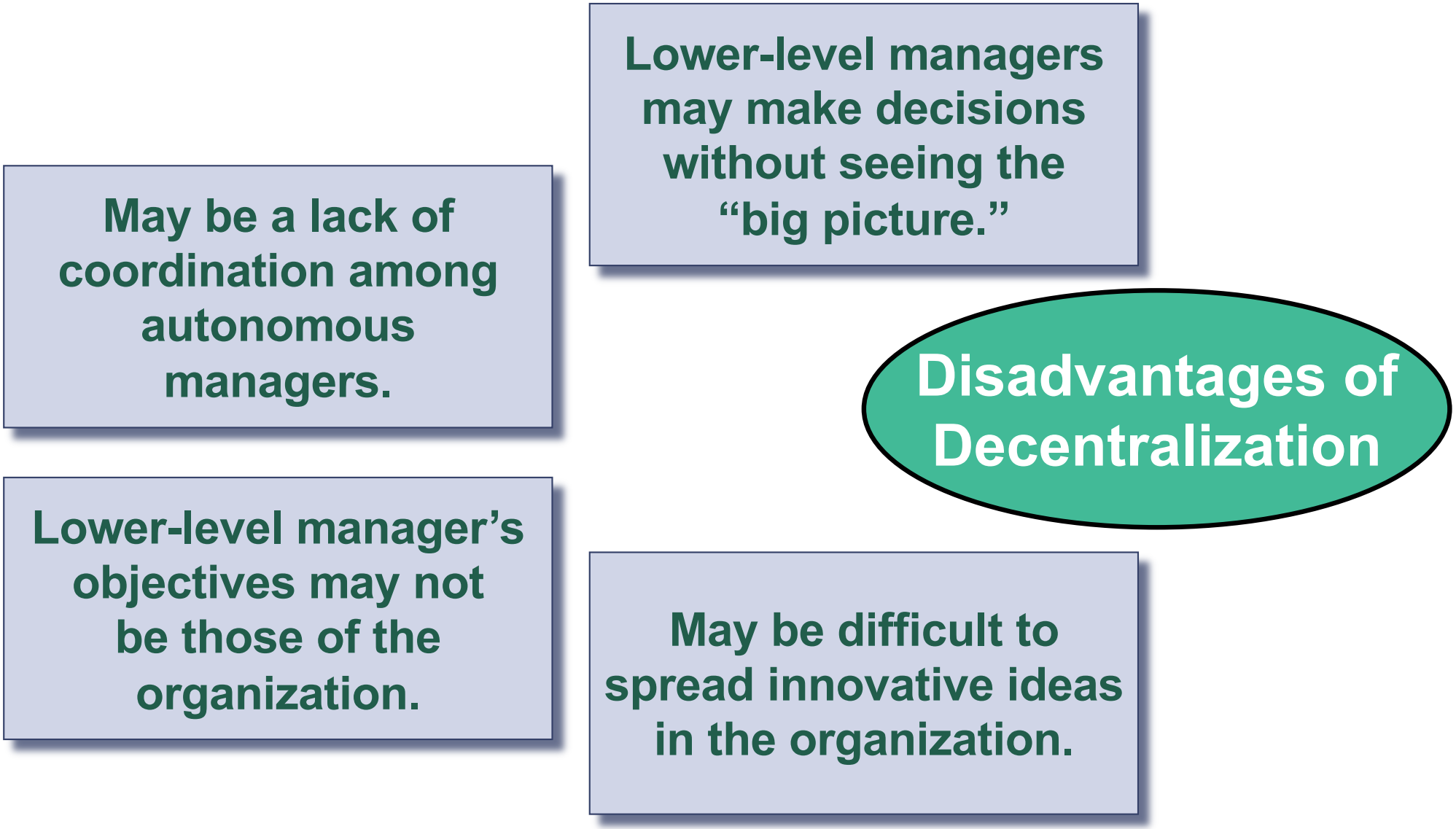
Benefits of Decentralization

Disadvantages of Decentralization

Decentralization in Organizations: Benefits



Decentralization in Organizations: Disadvantages



Responsibility Accounting

Managers are held responsible for those items—and *only* those items—that they can actually control to a significant extent.

A responsibility center is used for any part of an organization whose manager has control over and is accountable for cost, profit, or investments.

Responsibility centers

The three main type of RC are:

1. Cost centers

2. Profit centers

3. Investments centers

Cost Center

A segment whose manager has control over costs, but not over revenues or investment funds.

Profit Center

A segment whose manager has control over both costs and revenues, but no control over investment funds.

Revenues

- Sales
- Interest
- Other

Costs

- Mfg. costs
- Commissions
- Salaries
- Other

Investment Center

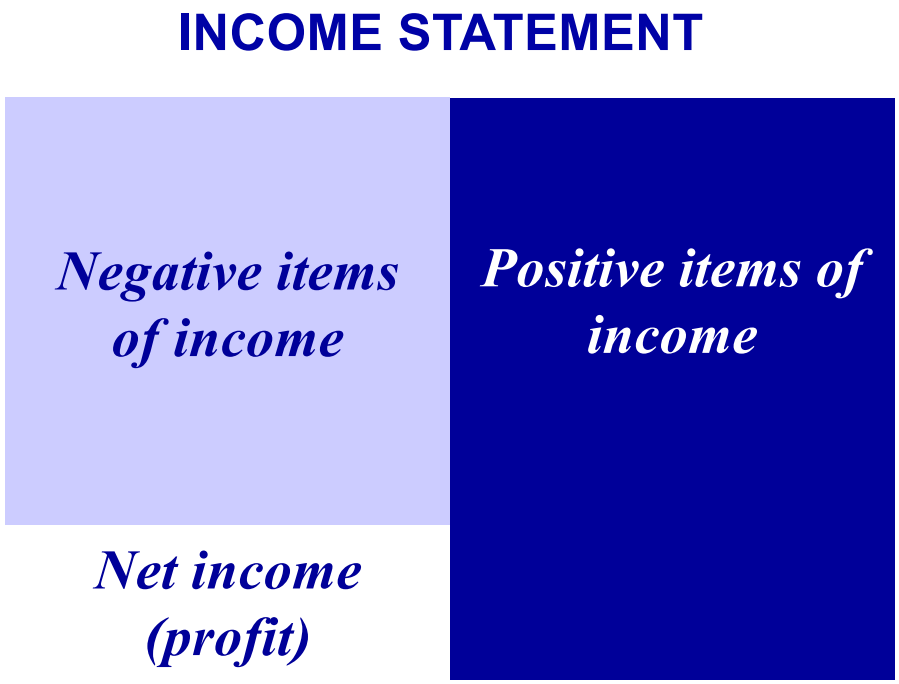
A segment whose manager has control over costs, revenues, and investments in operating assets.

Learning Objective 1

Compute return on investment (ROI) and show how changes in sales, expenses, and assets affect ROI.

Basics on financial statement

The FS is a mandatory set of documents prepared yearly by firms.
It includes two main accounting documents.

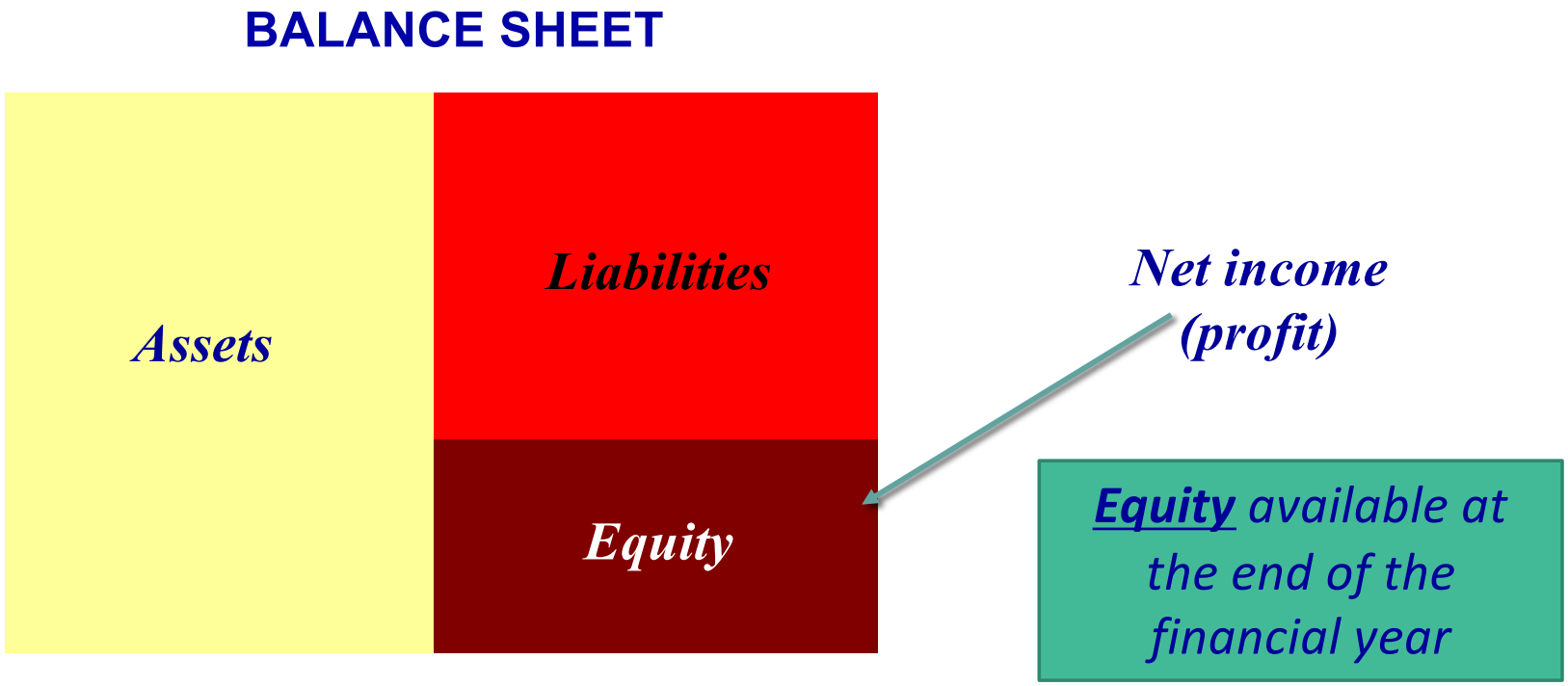


Net income *realized during the financial year*

Basics on financial statement

The FS is a mandatory set of documents prepared yearly by firms.

It includes two main accounting documents.



Basics on financial statement

To assess the financial health of companies, Financial Statement is usually «analysed» and the items of BS and IS are positioned within each document in order to obtain more information on the company and its performance.

Assets

Cash and cash equivalent: amounts in checking accounts and saving accounts + money

Account receivables or trade r.: amounts due from customers for goods or services that have already been delivered or sold

Inventories: goods kept in stock, available for sale

Property plant and equipment: assets which cannot be readily converted into cash (computers, cars, plants..)

Securities: bonds, stocks or share of other firms acquired (investment purposes)

Norton Balance Sheet – Part 1

NORTON CORPORATION Balance Sheets December 31		
	This Year	Last Year
Assets		
Current assets:		
Cash	$ 30,000	$ 20,000
Accounts receivable, net	20,000	17,000
Inventory	12,000	10,000
Prepaid expenses	3,000	2,000
Total current assets	65,000	49,000
Property and equipment:		
Land	165,000	123,000
Buildings and equipment, net	116,390	128,000
Total property and equipment	281,390	251,000
Total assets	$ 346,390	$ 300,000

Liabilities

Bank overdraft: withdraw fund (usually used by firms to pay suppliers)

Account payable or trade p.: amounts due to suppliers for goods or services that have already been acquired/received

Note payable: obligation to repay an amount of borrowed money at a particular future date

Severance provisions (employee benefit liabilities): obligation towards employees. It will be paid when the employee will finish his/her employment

Long term borrowings: obligations due to lenders (i.e. bank loan). Long term means that they are expected to be paid off beyond 12 months.

Equity

Share capital: amounts invested by owners of the company

Retained earnings (or reserves): the sum of net income (profit) over the life of the company which has not been distributed to owners in the form of dividend (dividends are distribution of profit)

Net income: difference between positive and negative items of income (profit or loss)

Norton Balance Sheet – Part 2

NORTON CORPORATION Balance Sheets December 31		
	This Year	Last Year
Liabilities and Stockholders' Equity		
Current liabilities:		
Accounts payable	$ 39,000	$ 40,000
Notes payable, short-term	3,000	2,000
Total current liabilities	42,000	42,000
Long-term liabilities:		
Notes payable, long-term	70,000	78,000
Total liabilities	112,000	120,000
Stockholders' equity:		
Common stock, $1 par value	27,400	17,000
Additional paid-in capital	158,100	113,000
Total paid-in capital	185,500	130,000
Retained earnings	48,890	50,000
Total stockholders' equity	234,390	180,000
Total liabilities and stockholders' equity	$ 346,390	$ 300,000

Par value = nominal value

APIC: money an investor pays above and beyond the par value

Norton Income Statement

Gross margin = revenues - cost of goods sold (only product not selling and admin)

Cost of financial resources

NORTON CORPORATION Income Statements For the Years Ended December 31		
	This Year	Last Year
Sales	$ 494,000	$ 450,000
Cost of goods sold	140,000	127,000
Gross margin	354,000	323,000
Operating expenses	270,000	249,000
Net operating income	84,000	74,000
Interest expense	7,300	8,000
Net income before taxes	76,700	66,000
Less income taxes (30%)	23,010	19,800
Net income	$ 53,690	$ 46,200

Return on Investment (ROI)

ROI compares the operating income produced by the firm to the operating assets.

It is a measure of profitability.

It measures the ability of the company to generate operating income by using its operating assets (**is the core business going well?)**

Return on Investment (ROI) Formula

Income before interest and taxes (EBIT)

$$\text{ROI} = \frac{\text{Net operating income}}{\text{Average operating assets}}$$

Cash, accounts receivable, inventory, plant and equipment, and other productive assets.

Understanding ROI

$$\text{ROI} = \frac{\text{Net operating income}}{\text{Average operating assets}}$$

$$\text{Margin} = \frac{\text{Net operating income}}{\text{Sales}}$$

$$\text{Turnover} = \frac{\text{Sales}}{\text{Average operating assets}}$$

$$\text{ROI} = \text{Margin} \times \text{Turnover}$$

Margin

$$\text{Margin} = \frac{\text{Net operating income}}{\text{Sales}}$$

The margin indicates a company's ability to earn an operative income (revenues - op. expenses) from sales.

Turnover

$$\text{Turnover} = \frac{\text{Sales}}{\text{Average operating assets}}$$

Turnover is a measure of the efficiency in the use of the firm's operating assets.

It expresses the number of times that the assets «are used» to produce sales (during a year).

Increasing ROI: An Example

Regal Company reports the following:

Net operating income	$ 30,000
Average operating assets	$ 200,000
Sales	$ 500,000
Operating expenses	$ 470,000

What is Regal Company's ROI?

$$\text{ROI} = \text{Margin} \times \text{Turnover}$$

$$\text{ROI} = \frac{\text{Net operating income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Average operating assets}}$$

Increasing ROI – An Example: Solution

$$\text{ROI} = \text{Margin} \times \text{Turnover}$$

$$\text{ROI} = \frac{\text{Net operating income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Average operating assets}}$$

$$\text{ROI} = \frac{\$30{,}000}{\$500{,}000} \times \frac{\$500{,}000}{\$200{,}000}$$

$$\text{ROI} = 6\% \times 2.5 = 15\%$$

Investing in Operating Assets to Increase Sales – An Example

Assume that Regal's manager invests in a $30,000 piece of equipment that increases sales by $35,000, while increasing operating expenses by $15,000.

Regal Company reports the following:

Net operating income	$ 50,000
Average operating assets	$ 230,000
Sales	$ 535,000
Operating expenses	$ 485,000

Let's calculate the new ROI.

Investing in Operating Assets to Increase Sales – An Example: Solution

$$\text{ROI} = \text{Margin} \times \text{Turnover}$$

$$\text{ROI} = \frac{\text{Net operating income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Average operating assets}}$$

$$\text{ROI} = \frac{\$50{,}000}{\$535{,}000} \times \frac{\$535{,}000}{\$230{,}000}$$

$$\text{ROI} = 9.35\% \times 2.33 = 21.8\%$$

ROI increased from 15% to 21.8%.

Criticisms of ROI

In the absence of the balanced scorecard, management may not know how to increase ROI.

Managers often inherit many committed costs over which they have no control.

Managers evaluated on ROI may reject profitable investment opportunities.

Learning Objective 2

Compute residual income and explain its strengths and weaknesses.

Residual Income – Another Measure of Performance

Residual Income is net operating income above some minimum return on operating assets.

Calculating Residual Income

$$\text{Residual income} = \text{Net operating income} - \left(\text{Average operating assets} \times \text{Minimum required rate of return}\right)$$

This computation differs from ROI.

ROI measures net operating income earned relative to the investment in average operating assets.

Residual income measures net operating income earned less the minimum required return on average operating assets.

Residual Income – An Example

The Retail Division of Zephyr, Inc. has average operating assets of $100,000 and is required to earn a return of 20% on these assets.

In the current period, the division earns $30,000.

Let's calculate residual income.

Residual Income – An Example: Solution

Operating assets		$ 100,000
Required rate of return	×	20%
Minimum required return		$ 20,000

Actual income	$ 30,000
Minimum required return	(20,000)
Residual income	$ 10,000

Motivation and Residual Income

Residual income encourages managers to make profitable investments that would be rejected by managers using ROI.

Quick Check 1

Redmond Awnings, a division of Wrap-up Corp., has a net operating income of $60,000 and average operating assets of $300,000. The required rate of return for the company is 15%. What is the division's ROI?

a. 25%

b. 5%

c. 15%

d. 20%

Quick Check 2

Redmond Awnings, a division of Wrap-up Corp., has a net operating income of $60,000 and average operating assets of $300,000. If the manager of the division is evaluated based on ROI, will she want to make an investment of $100,000 that would generate additional net operating income of $18,000 per year?

a. Yes

b. No

Quick Check 3

The company's required rate of return is 15%. Would the company want the manager of the Redmond Awnings division to make an investment of $100,000 that would generate additional net operating income of $18,000 per year?

a. Yes

b. No

Quick Check 4

Redmond Awnings, a division of Wrap-up Corp., has a net operating income of $60,000 and average operating assets of $300,000. The required rate of return for the company is 15%. What is the division's residual income?

a. $240,000

b. $ 45,000

c. $ 15,000

d. $ 51,000

Learning Objective 3

Determine the range, if any, within which a negotiated transfer price should fall.

Key Concepts/Definitions

A transfer price is the price charged when one segment of a company provides goods or services to another segment of the company.

The fundamental objective in setting transfer prices is to motivate managers to act in the best interests of the overall company.

Three Primary Approaches

There are three primary approaches to setting transfer prices:

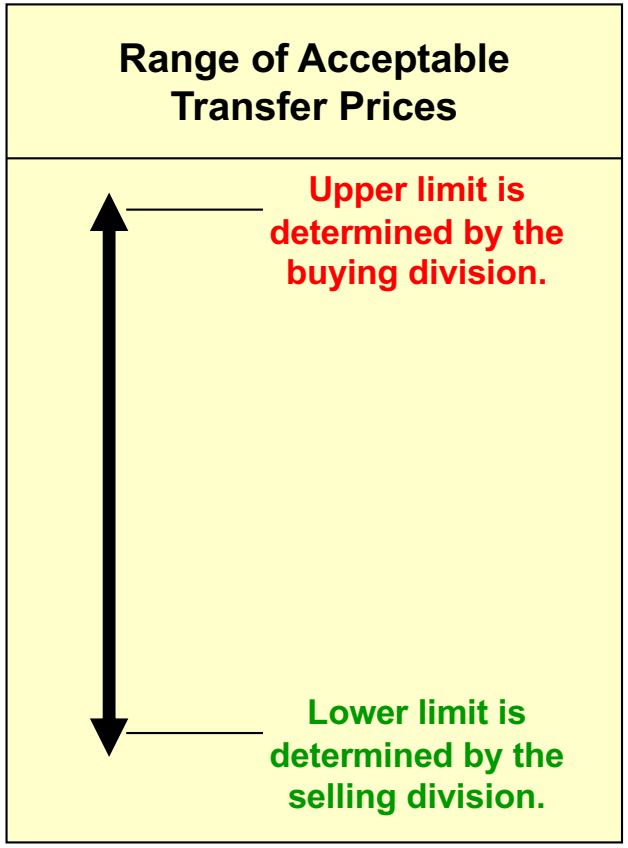
1. **Negotiated transfer prices**
2. **Set transfer prices at cost using either variable cost or full cost**
3. **Transfers at market price**

Negotiated Transfer Prices

A negotiated transfer price results from discussions between the selling and buying divisions.

Advantages of negotiated transfer prices:

1. They preserve the autonomy of the divisions, which is consistent with the spirit of decentralization.
2. The managers negotiating the transfer price are likely to have much better information about the potential costs and benefits of the transfer than others in the company.



Grocery Storehouse – Part 1

Assume the information as shown with respect to West Coast Plantations and Grocery Mart (both companies are owned by Grocery Storehouse).

West Coast Plantations:			
Naval orange harvest capactiy per month		10,000	crates
Variable cost per crate of naval oranges	$	10	per crate
Fixed costs per month	$	100,000	
Selling price of navel oranges on the outside market	$	25	per crate
Grocery Mart:			
Purchase price of current naval oranges	$	20	per crate
Monthly sales of naval oranges		1,000	crates

Grocery Storehouse – Part 2

The selling division's (West Coast Plantations) lowest acceptable transfer price is calculated as:

$$\text{Transfer Price} \geq \text{Variable cost per unit} + \frac{\text{Total contribution margin on lost sales}}{\text{Number of units transferred}}$$

Let's calculate the lowest and highest acceptable transfer prices under three scenarios.

The buying division's (Grocery Mart) highest acceptable transfer price is calculated as:

$$\text{Transfer Price} \leq \text{Cost of buying from outside supplier}$$

If an outside supplier does not exist, the highest acceptable transfer price is calculated as:

$$\text{Transfer Price} \leq \text{Profit to be earned per unit sold (not including the transfer price)}$$

Grocery Storehouse – Part 3

If West Coast Plantations has sufficient idle capacity (3,000 crates) to satisfy Grocery Mart's demands (1,000 crates), without sacrificing sales to other customers, then the lowest and highest possible transfer prices are computed as follows:

Selling division's lowest possible transfer price:

$$\text{Transfer Price} \geq \$10 + \frac{\$ -}{1{,}000} = \$10$$

Buying division's highest possible transfer price:

$$\text{Transfer Price} \leq \text{Cost of buying from outside supplier} = \$\ 20$$

Therefore, the range of acceptable transfer prices is $10 – $20.

Grocery Storehouse – Part 4

If West Coast Plantations has no idle capacity (0 crates) and must sacrifice other customer orders (1,000 crates) to meet Grocery Mart's demands (1,000 crates), then the lowest and highest possible transfer prices are computed as follows:

Selling division's lowest possible transfer price:

$$\text{Transfer Price} \geq \$\ 10 + \frac{(\$25 - \$10) \times 1{,}000}{1{,}000} = \$\ 25$$

Buying division's highest possible transfer price:

$$\text{Transfer Price} \leq \text{Cost of buying from outside supplier} = \$\ 20$$

Therefore, there is no range of acceptable transfer prices.

Grocery Storehouse – Part 5

If West Coast Plantations has some idle capacity (500 crates) and must sacrifice other customer orders (500 crates) to meet Grocery Mart's demands (1,000 crates), then the lowest and highest possible transfer prices are computed as follows:

Selling division's lowest possible transfer price:

$$\text{Transfer Price} \geq \$\,10 + \frac{(\$25 - \$10) \times 500}{1{,}000} = \$\,17.50$$

Buying division's highest possible transfer price:

$$\text{Transfer Price} \leq \text{Cost of buying from outside supplier} = \$\,20$$

Therefore, the range of acceptable transfer prices is $17.50 – $20.00.

Evaluation of Negotiated Transfer Prices

If a transfer within a company would result in higher overall profits for the company, there is always a range of transfer prices within which both the selling and buying divisions would have higher profits if they agree to the transfer.

If managers are pitted against each other rather than against their past performance or reasonable benchmarks, a noncooperative atmosphere is almost guaranteed.

Given the disputes that often accompany the negotiation process, most companies rely on some other means of setting transfer prices.

Three Primary Approaches

There are three primary approaches to setting transfer prices:

1. **Negotiated transfer prices**
2. **Set transfer prices at cost using either variable cost or full cost**
3. **Transfers at market price**

Transfers at the Cost to the Selling Division

Many companies set transfer prices at either the variable cost or full cost incurred by the selling division.

Drawbacks of this approach include:

1. **Using full cost as a transfer price can lead to suboptimization.**
2. **The selling division will never show a profit on any internal transfer.**
3. **Cost-based transfer prices do not provide incentives to control costs.**

Three Primary Approaches

There are three primary approaches to setting transfer prices:

1. **Negotiated transfer prices**
2. **Set transfer prices at cost using either variable cost or full cost**
3. **Transfers at market price**

Transfers at Market Price

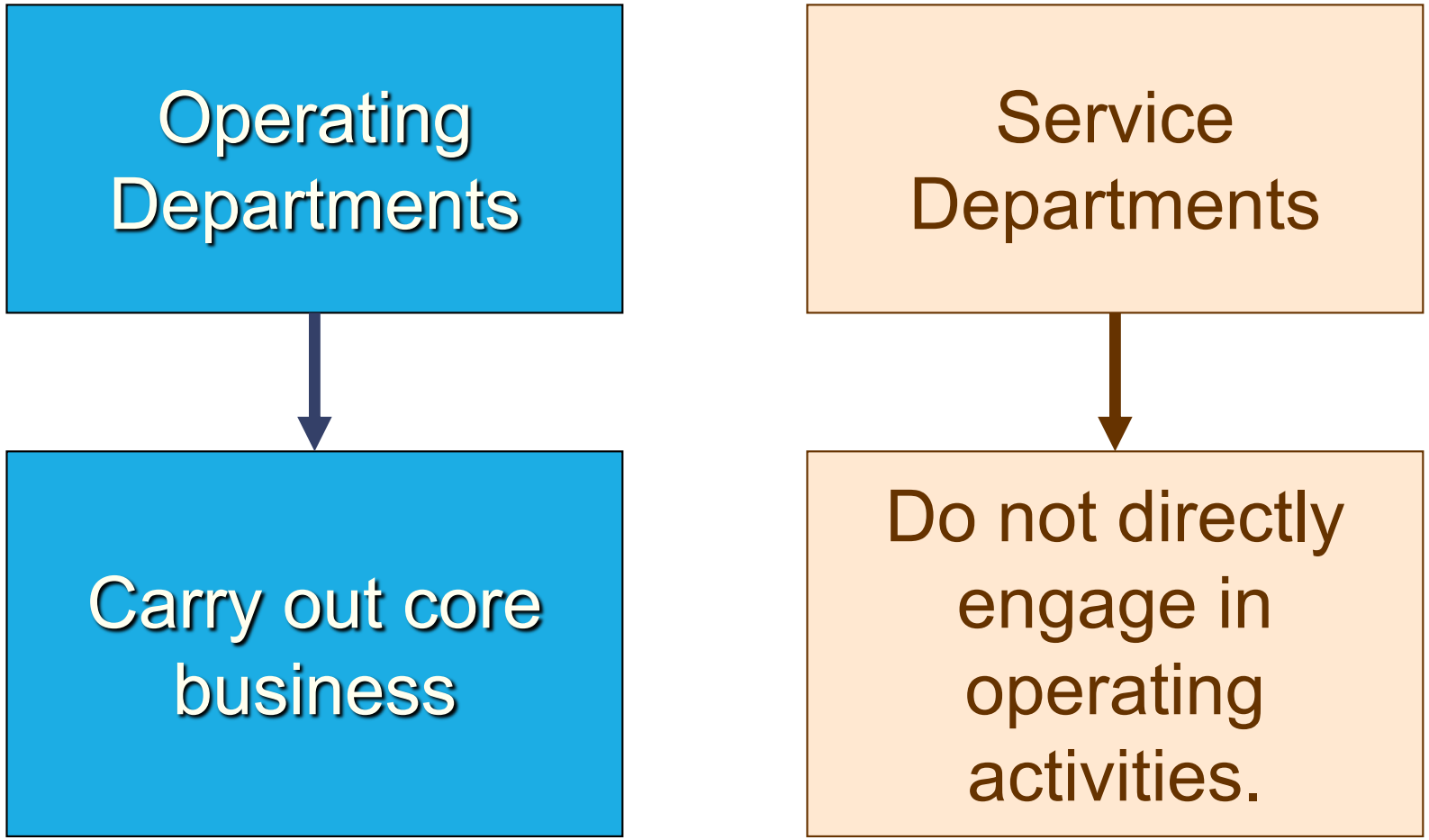
A market price (i.e., the price charged for an item on the open market) is often regarded as the best approach to the transfer pricing problem.

1. **A market price approach works best when the product or service is sold in its present form to outside customers and the selling division has no idle capacity.**
2. **A market price approach does not work well when the selling division has idle capacity.**

Learning Objective 4

Charge operating departments for services provided by service departments.

Operating vs. Service Departments



Reasons for Charging Service Department Costs

Service department costs are charged to operating departments for a variety of reasons including:

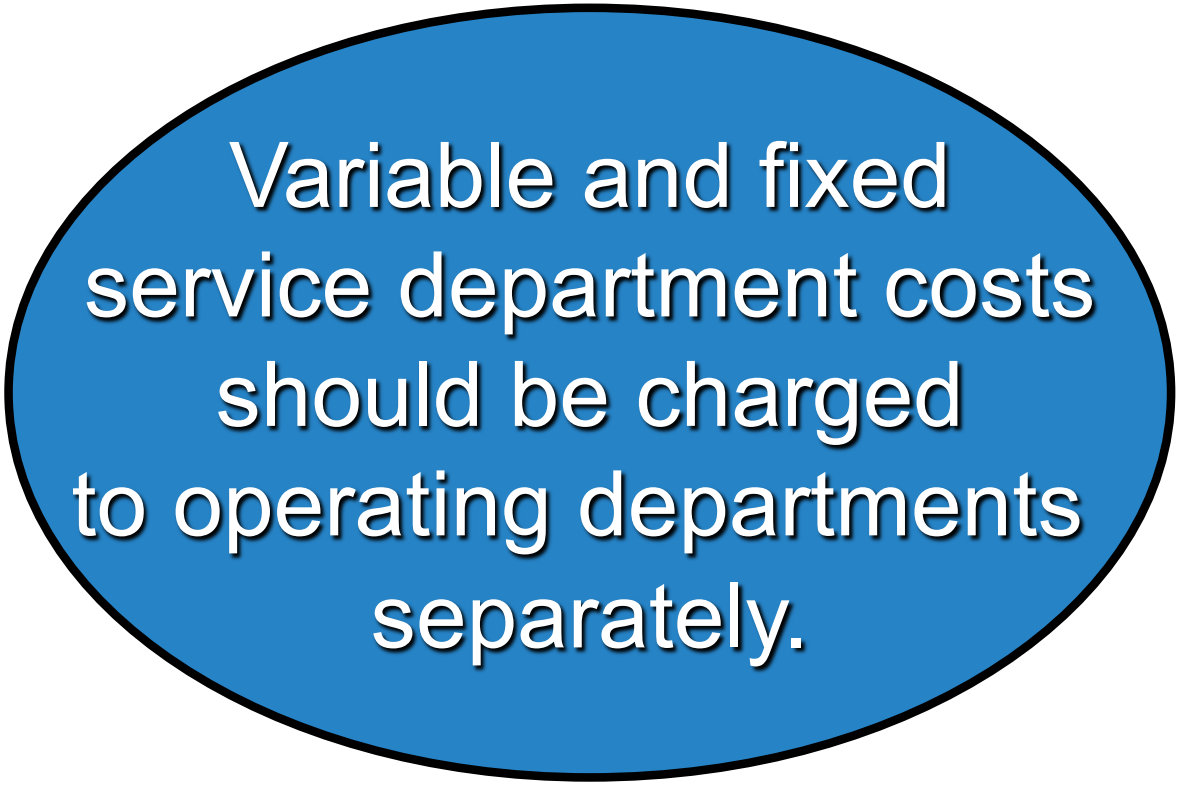
To encourage operating departments to wisely use service department resources.

To provide operating departments with more complete cost data for making decisions.

To help measure the profitability of operating departments.

To create an incentive for service departments to operate efficiently.

Charging Costs by Behavior – Part 1

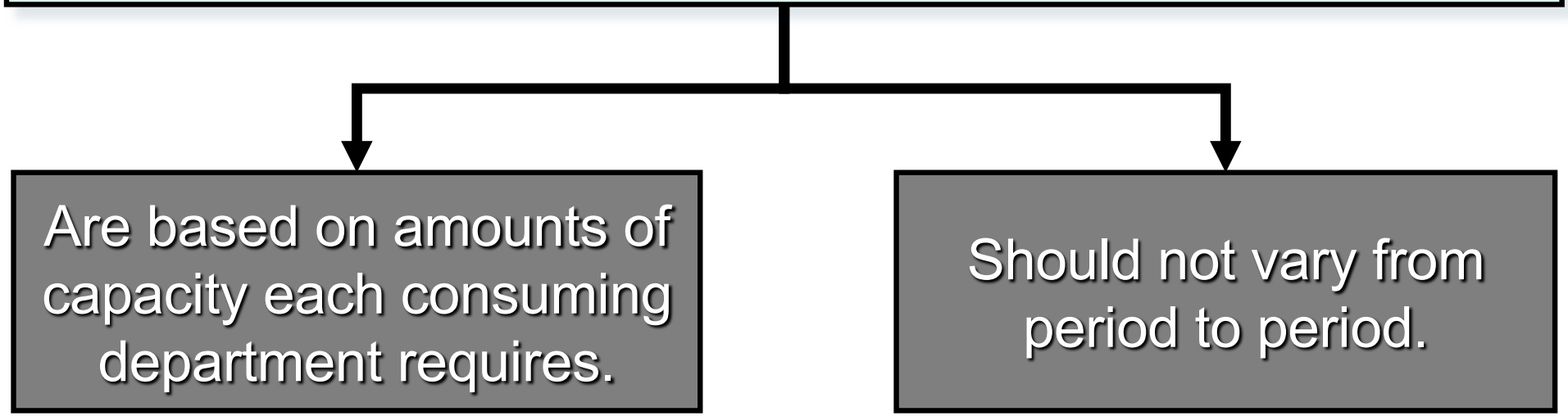


Charging Costs by Behavior – Part 2

Variable service department costs should be charged to consuming departments according to whatever activity (or cost driver) causes the incurrence of the cost.

Charging Costs by Behavior – Part 3

Fixed service department costs are assigned to consuming departments in predetermined amounts that are based on the consuming department's peak-period or long-run average servicing needs.



Should Actual or Budgeted Costs Be Charged?

Budgeted variable and fixed service department costs should be charged to operating departments.

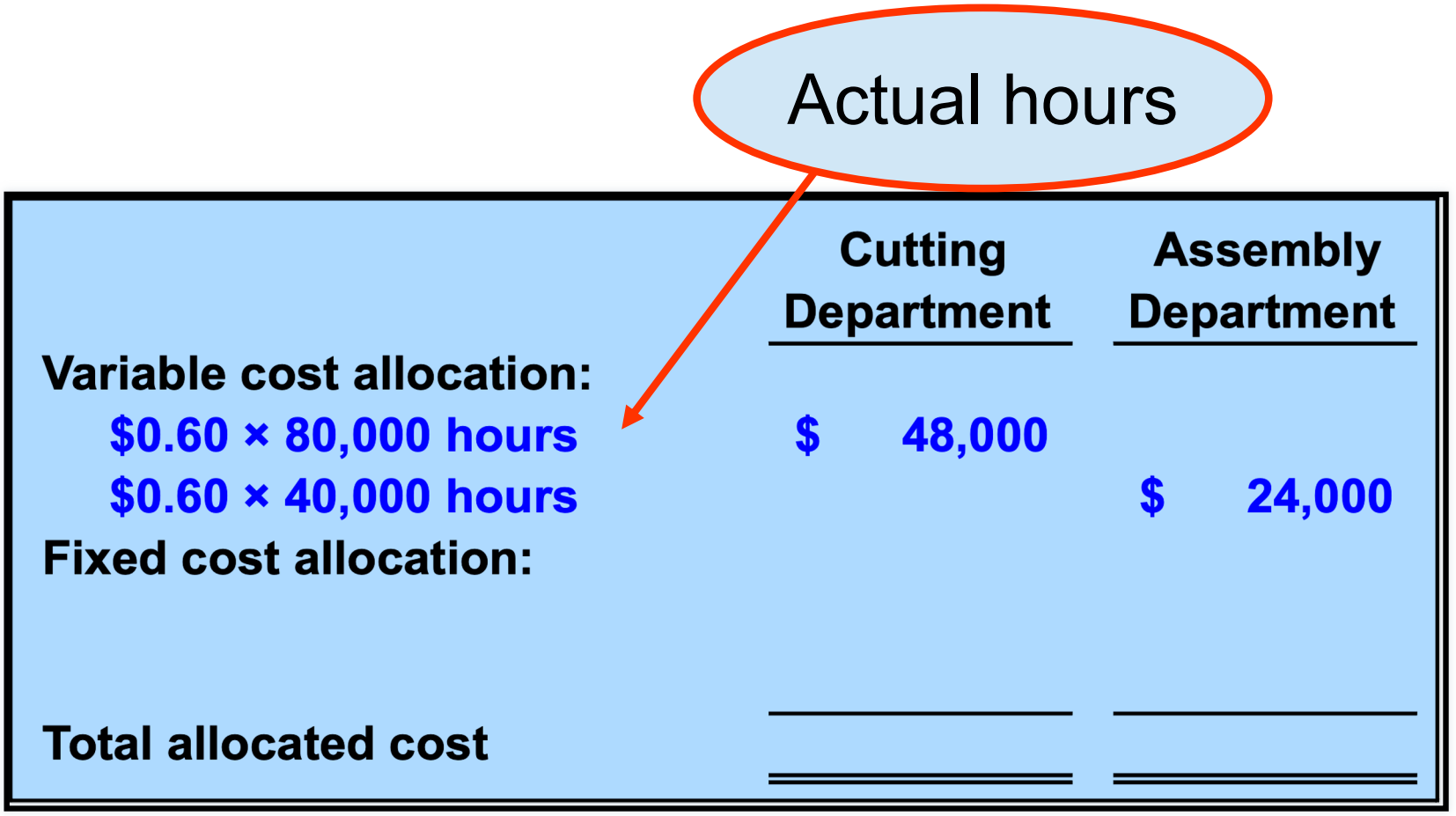
Sipco – An Example

Sipco has a maintenance department and two operating departments: Cutting and Assembly. **Variable** maintenance **costs** are budgeted at $0.60 per machine hour. **Fixed** maintenance **costs** are budgeted at $200,000 per year. Data relating to the current year are:

Operating Departments	Percent of Peak-Period Capacity Required	Hours Planned	Hours Used
Cutting	60%	75,000	80,000
Assembly	40%	50,000	40,000
Total hours	100%	125,000	120,000

Allocate maintenance costs to the two operating departments.

Sipco – End of the Year – Part 1



Sipco – End of the Year – Part 2

Actual hours

	Cutting Department	Assembly Department
Variable cost allocation:		
$0.60 × 80,000 hours	$ 48,000	
$0.60 × 40,000 hours		$ 24,000
Fixed cost allocation:		
60% × $200,000	120,000	
40% × $200,000		80,000
Total allocated cost	$ 168,000	$ 104,000

Percent of peak-period capacity.

Quick Check 1

Foster City has an ambulance service that is used by the two public hospitals in the city. Variable ambulance costs are budgeted at $4.20 per mile. Fixed ambulance costs are budgeted at $120,000 per year. Data relating to the current year are:

Hospitals	Percent of Peak-Period Capacity Required	Miles Planned	Miles Used
Mercy	45%	15,000	16,000
Northside	55%	17,000	17,500
Total	100%	32,000	33,500

Quick Check 1a

How much ambulance service cost will be allocated to Mercy Hospital at the end of the year?

a. $121,200

b. $254,400

c. $139,500

d. $117,000

Autos R Us – An Example

Autos R Us has one service department and three sales departments, New Cars, Used Cars, and Car Parts. The service department costs total $80,000 for both years in the example. Contrary to good practice, Autos R Us allocates the service department costs based on sales.

Autos R Us – First-year Allocation

	Departments			
	New	Used	Parts	Total
Sales by department	$ 1,500,000	$ 900,000	$ 600,000	$ 3,000,000
Percentage of total sales	50%	30%	20%	100%
Allocation of service department costs	$ 40,000	$ 24,000	$ 16,000	$ 80,000

$1,500,000 ÷ $3,000,000

50% of $80,000

In the next year, the manager of the New Cars department increases sales by $500,000. Sales in the other departments are unchanged. Let's allocate the $80,000 service department cost for the second year given the sales increase.

Autos R Us – Second-year Allocation

	Departments			
	New	Used	Parts	Total
Sales by department	$ 2,000,000	$ 900,000	$ 600,000	$ 3,500,000
Percentage of total sales	57%	26%	17%	100%
Allocation of service department costs	$ 45,714	$ 20,571	$ 13,715	$ 80,000

$2,000,000 ÷ $3,500,000

57% of $80,000

If you were the manager of the New Cars department, you would likely complain about the increased service department costs allocated to your department.